# SIMATS ENGINEERING

## ASSESSMENT TOOL 2

**COURSE NAME:** **Database Management Systems**

**COURSE CODE:** **CSA05**

---

**COURSE OUTCOME COVERED**

**CO3:** *Analyze relational databases using functional dependencies, normalization (1NF to 5NF), and key constraints to identify redundancy and ensure data integrity.* (BL2, BL3)

Activity Tool : Case Study (Medium)

Assessment Tool Weightage: 30%

---

| **QUESTIONS** | | Total Marks: 50 | |
|---|---|---|---|
| **Q.No** | **Question** | **Bloom's Level** | **Marks** |
| **1** | Analyze the case: An online shopping platform stores Order(OrderID, CustID, CustName, ProdID, ProdName, Qty, Price). Identify FDs and normalize to 3NF. | BL2 – Understand | **5** |
| **2** | A hospital maintains Patient(PID, PName, DocID, DocName, Specialization, WardNo, WardName). Identify anomalies and normalize to BCNF. | BL3 – Apply | **5** |

| | | | |
|---|---|---|---|
| **3** | Case study: A university stores Enrollment(SID, SName, CourseID, CourseName, Faculty, Grade). Analyze FDs and apply 2NF decomposition. | BL3 – Apply | **5** |
| **4** | Given the airline booking case: Booking(FlightNo, PassID, PassName, Seat, DeptCity, ArrCity, Date). Normalize up to BCNF. | BL3 – Apply | **5** |
| **5** | Analyze an HR system schema for a multinational company. Identify all functional and transitive dependencies and normalize to 3NF. | BL2 – Understand | **5** |
| **6** | Case: Library(MemberID, MemberName, BookID, Title, Author, Genre, BorrowDate). Identify MVDs and decompose to 4NF. | BL3 – Apply | **5** |
| **7** | Analyze a telecom billing schema and identify all candidate keys, primary keys, and functional dependencies before normalization. | BL2 – Understand | **5** |
| **8** | A retail inventory system has Product(PID, PName, SupplierID, SupplierName, Category, Price, Warehouse). Apply complete normalization. | BL3 – Apply | **5** |
| **9** | Study the given poorly normalized schema for a School ERP and propose a fully normalized design up to BCNF with justification. | BL3 – Apply | **5** |
| **10** | Given a movie streaming platform schema, identify join dependencies and decompose to 5NF with lossless-join verification. | BL3 – Apply | **5** |

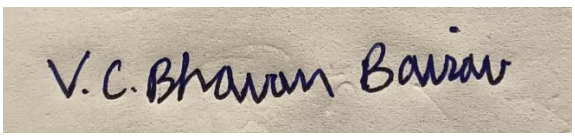
---

***Code of Conduct:***

*I V.C.BHAVAN BAIRAV (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***



**RUBRICS FOR CALCULATION:**

| Criteria | Level 4 – Exemplary | Level 3 – Proficient | Level 2 – Developing | Level 1 – Beginning |
|---|---|---|---|---|
| Understanding of Case | Thorough understanding; clearly identifies all key issues | Good understanding with most issues identified | Basic understanding; some issues missed | Poor understanding; problem not identified correctly |
| Application of Concepts | Effectively applies relevant concepts to analyze the case deeply | Applies appropriate concepts with minor gaps | Limited application of concepts | Fails to apply relevant concepts |
| Analysis | In-depth analysis with strong reasoning and insights | Good analysis with logical reasoning | Basic analysis with limited depth | Weak or no analysis; lacks reasoning |
| Solution Design | Innovative, feasible solutions with strong justification | Logical solutions with adequate justification | Basic solutions with weak justification | Incorrect or no solution provided |
| Clarity | Clear, well-structured, and easy to understand | Organized and understandable presentation | Limited clarity and structure | Poor clarity; difficult to understand |

1) Analyze the case: An online shopping platform stores Order(OrderID, CustID, CustName, ProdID, ProdName, Qty, Price). Identify FDs and normalize to 3NF.

**AIM:**

To identify the functional dependencies in the given Order relation and normalize it up to Third Normal Form (3NF).

**GIVEN RELATION:**

```
Database changed
mysql> CREATE TABLE ONLINE_ORDER (
    ->     OrderID INT,
    ->     CustID INT,
    ->     CustName VARCHAR(50),
    ->     ProdID INT,
    ->     ProdName VARCHAR(50),
    ->     Qty INT,
    ->     Price DECIMAL(10,2),
    ->     PRIMARY KEY (OrderID, ProdID)
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql> INSERT INTO ONLINE_ORDER VALUES
    -> (1001, 501, 'Arun', 101, 'Laptop', 1, 55000.00),
    -> (1001, 501, 'Arun', 102, 'Mouse', 2, 800.00),
    -> (1002, 502, 'Meena', 103, 'Keyboard', 1, 1500.00),
    -> (1003, 503, 'Kiran', 101, 'Laptop', 1, 55000.00);
Query OK, 4 rows affected (0.02 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM ONLINE_ORDER;
+---------+--------+----------+--------+----------+------+----------+
| OrderID | CustID | CustName | ProdID | ProdName | Qty  | Price    |
+---------+--------+----------+--------+----------+------+----------+
|    1001 |    501 | Arun     |    101 | Laptop   |    1 | 55000.00 |
|    1001 |    501 | Arun     |    102 | Mouse    |    2 |   800.00 |
|    1002 |    502 | Meena    |    103 | Keyboard |    1 |  1500.00 |
|    1003 |    503 | Kiran    |    101 | Laptop   |    1 | 55000.00 |
+---------+--------+----------+--------+----------+------+----------+
4 rows in set (0.01 sec)
```

**FUNCTIONAL DEPENDENCIES**

- OrderID → CustID
- CustID → CustName
- ProdID → ProdName, Price
- (OrderID, ProdID) → Qty

**CANDIDATE KEY**

- **(**OrderID, ProdID)

Since an order can contain multiple products, both OrderID and ProdID are required to uniquely identify the quantity.

**IDENTIFY PARTIAL DEPENDENCIES**

The candidate key is:

- (OrderID, ProdID)

But:

- OrderID → CustID
- CustID → CustName
- ProdID → ProdName, Price

CustID, CustName, ProdName, and Price do not depend on the complete composite key.

Therefore, partial dependencies exist.

**3NF DECOMPOSITION**

MySQL 8.0 Command Line Client

```
mysql> CREATE TABLE CUSTOMER (
    ->     CustID INT PRIMARY KEY,
    ->     CustName VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE PRODUCT (
    ->     ProdID INT PRIMARY KEY,
    ->     ProdName VARCHAR(50),
    ->     Price DECIMAL(10,2)
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql> CREATE TABLE ORDERS (
    ->     OrderID INT PRIMARY KEY,
    ->     CustID INT,
    ->     FOREIGN KEY (CustID) REFERENCES CUSTOMER(CustID)
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql> CREATE TABLE ORDER_DETAILS (
    ->     OrderID INT,
    ->     ProdID INT,
    ->     Qty INT,
    ->     PRIMARY KEY (OrderID, ProdID),
    ->     FOREIGN KEY (OrderID) REFERENCES ORDERS(OrderID),
    ->     FOREIGN KEY (ProdID) REFERENCES PRODUCT(ProdID)
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql>
```

```
mysql> INSERT INTO CUSTOMER VALUES
    -> (501, 'Arun'),
    -> (502, 'Meena'),
    -> (503, 'Kiran');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> INSERT INTO PRODUCT VALUES
    -> (101, 'Laptop', 55000.00),
    -> (102, 'Mouse', 800.00),
    -> (103, 'Keyboard', 1500.00);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> INSERT INTO ORDERS VALUES
    -> (1001, 501),
    -> (1002, 502),
    -> (1003, 503);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> INSERT INTO ORDER_DETAILS VALUES
    -> (1001, 101, 1),
    -> (1001, 102, 2),
    -> (1002, 103, 1),
    -> (1003, 101, 1);
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

MySQL 8.0 Command Line Client

```
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM CUSTOMER;
+--------+----------+
| CustID | CustName |
+--------+----------+
|    501 | Arun     |
|    502 | Meena    |
|    503 | Kiran    |
+--------+----------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM PRODUCT;
+--------+----------+----------+
| ProdID | ProdName | Price    |
+--------+----------+----------+
|    101 | Laptop   | 55000.00 |
|    102 | Mouse    |   800.00 |
|    103 | Keyboard |  1500.00 |
+--------+----------+----------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM ORDERS;
+---------+--------+
| OrderID | CustID |
+---------+--------+
|    1001 |    501 |
|    1002 |    502 |
|    1003 |    503 |
+---------+--------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM ORDER_DETAILS;
+---------+--------+------+
| OrderID | ProdID | Qty  |
+---------+--------+------+
|    1001 |    101 |    1 |
|    1001 |    102 |    2 |
|    1002 |    103 |    1 |
|    1003 |    101 |    1 |
+---------+--------+------+
4 rows in set (0.00 sec)
```

**RESULT:**

Thus, the functional dependencies in the online shopping Order relation were identified and the relation was successfully normalized into 3NF, reducing redundancy and preventing insertion, deletion, and update anomalies. The decomposition follows the same 3NF structure shown in the assessment material.

2) A hospital maintains Patient(PID, PName, DocID, DocName, Specialization, WardNo, WardName). Identify anomalies and normalize to BCNF.

**AIM:**

To analyze the given hospital database relation, identify the functional dependencies and anomalies, and decompose the relation into Boyce-Codd Normal Form (BCNF) to reduce redundancy and improve data consistency.

**GIVEN RELATION:**

- Patient(PID, PName, DocID, DocName, Specialization, WardNo, WardName)

**FUNCTIONAL DEPENDENCIES:**

- PID → PName, DocID, WardNo
- DocID → DocName, Specialization
- WardNo → WardName

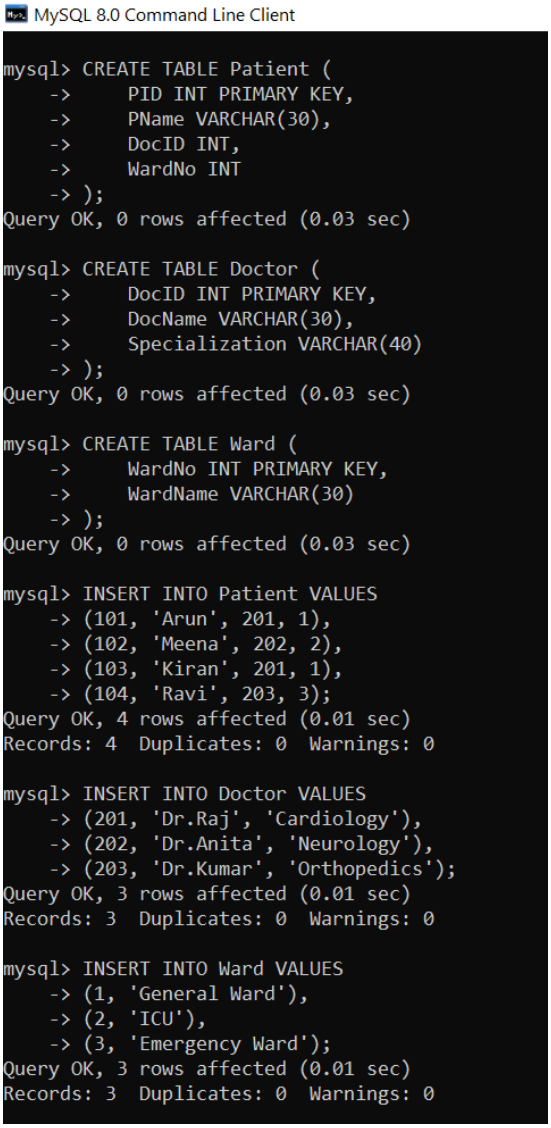
**ANOMALIES:**

- **Update anomaly:** If a doctor's specialization changes, it may have to be changed in multiple patient records.
- **Insertion anomaly:** A new doctor cannot be stored unless a patient is assigned to that doctor.

- **Deletion anomaly:** Deleting the last patient of a doctor may remove information about that doctor.

**BCNF DECOMPOSITION:**

- Patient(PID, PName, DocID, WardNo)
- Doctor(DocID, DocName, Specialization)
- Ward(WardNo, WardName)



```
mysql> SELECT * FROM Patient;
+-----+-------+-------+--------+
| PID | PName | DocID | WardNo |
+-----+-------+-------+--------+
| 101 | Arun  |   201 |      1 |
| 102 | Meena |   202 |      2 |
| 103 | Kiran |   201 |      1 |
| 104 | Ravi  |   203 |      3 |
+-----+-------+-------+--------+
4 rows in set (0.00 sec)

mysql> SELECT * FROM Doctor;
+-------+----------+----------------+
| DocID | DocName  | Specialization |
+-------+----------+----------------+
|   201 | Dr.Raj   | Cardiology     |
|   202 | Dr.Anita | Neurology      |
|   203 | Dr.Kumar | Orthopedics    |
+-------+----------+----------------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM Ward;
+--------+----------------+
| WardNo | WardName       |
+--------+----------------+
|      1 | General Ward   |
|      2 | ICU            |
|      3 | Emergency Ward |
+--------+----------------+
3 rows in set (0.00 sec)
```

**JUSTIFICATION**

Each determinant becomes a key in its respective relation:

- PID is the key of Patient.
- DocID is the key of Doctor.
- WardNo is the key of Ward.

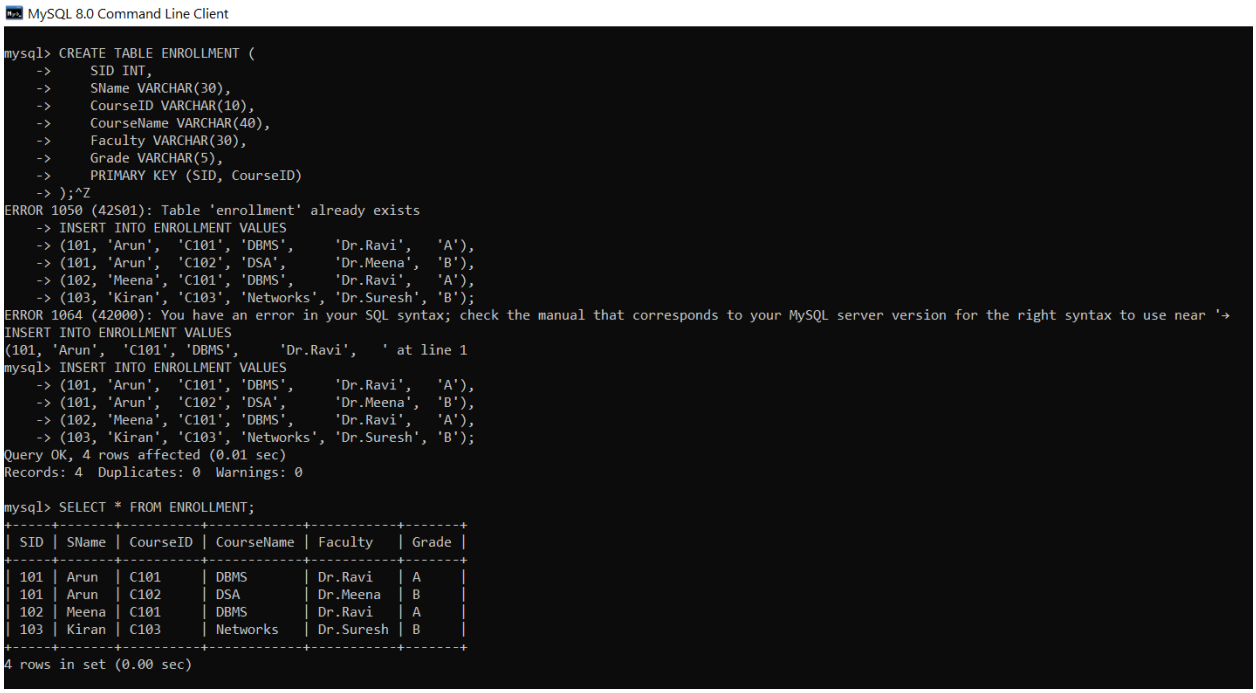
Therefore, the relations satisfy **BCNF**.

3) Case study: A university stores Enrollment (SID, SName, CourseID, CourseName, Faculty, Grade). Analyze FDs and apply 2NF decomposition.

**AIM:**

To identify the functional dependencies in the Enrollment relation and decompose it into Second Normal Form (2NF) by removing partial dependencies.

**GIVEN RELATION:**




**FUNCTIONAL DEPENDENCIES**

- SID → SName
- CourseID → CourseName, Faculty
- (SID, CourseID) → Grade

## CANDIDATE KEY

- (SID, CourseID)

## CREATE 2NF TABLES

```
mysql> CREATE TABLE STUDENT (
    ->     SID INT PRIMARY KEY,
    ->     SName VARCHAR(30)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE COURSE (
    ->     CourseID VARCHAR(10) PRIMARY KEY,
    ->     CourseName VARCHAR(40),
    ->     Faculty VARCHAR(30)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE ENROLLMENT_2NF (
    ->     SID INT,
    ->     CourseID VARCHAR(10),
    ->     Grade VARCHAR(5),
    ->     PRIMARY KEY (SID, CourseID),
    ->     FOREIGN KEY (SID) REFERENCES STUDENT(SID),
    ->     FOREIGN KEY (CourseID) REFERENCES COURSE(CourseID)
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql> INSERT INTO STUDENT VALUES
    -> (101, 'Arun'),
    -> (102, 'Meena'),
    -> (103, 'Kiran');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> INSERT INTO COURSE VALUES
    -> ('C101', 'DBMS', 'Dr.Ravi'),
    -> ('C102', 'DSA', 'Dr.Meena'),
    -> ('C103', 'Networks', 'Dr.Suresh');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> INSERT INTO ENROLLMENT_2NF VALUES
    -> (101, 'C101', 'A'),
    -> (101, 'C102', 'B'),
    -> (102, 'C101', 'A'),
    -> (103, 'C103', 'B');
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM STUDENT;
+-----+-------+
| SID | SName |
+-----+-------+
| 101 | Arun  |
| 102 | Meena |
| 103 | Kiran |
+-----+-------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM COURSE;
+----------+------------+-----------+
| CourseID | CourseName | Faculty   |
+----------+------------+-----------+
| C101     | DBMS       | Dr.Ravi   |
| C102     | DSA        | Dr.Meena  |
| C103     | Networks   | Dr.Suresh |
+----------+------------+-----------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM ENROLLMENT_2NF;
+-----+----------+-------+
| SID | CourseID | Grade |
+-----+----------+-------+
| 101 | C101     | A     |
| 101 | C102     | B     |
| 102 | C101     | A     |
| 103 | C103     | B     |
+-----+----------+-------+
4 rows in set (0.00 sec)
```

**RESULT:**

Thus, the Enrollment relation was successfully decomposed into 2NF by removing the partial dependencies. This matches the decomposition specified for Q3 in your assessment.

4) Given the airline booking case: Booking(FlightNo, PassID, PassName, Seat, DeptCity, ArrCity, Date). Normalize up to BCNF.

**AIM:**

To identify the functional dependencies in the airline Booking relation and normalize it up to Boyce-Codd Normal Form (BCNF).

**GIVEN RELATION:**

```
mysql> CREATE TABLE AIRLINE_BOOKING (
    ->     FlightNo VARCHAR(10),
    ->     PassID VARCHAR(10),
    ->     PassName VARCHAR(30),
    ->     Seat VARCHAR(10),
    ->     DeptCity VARCHAR(30),
    ->     ArrCity VARCHAR(30),
    ->     BookingDate DATE,
    ->     PRIMARY KEY (FlightNo, PassID)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO AIRLINE_BOOKING VALUES
    -> ('F101','P101','Arun','12A','Chennai','Delhi','2026-08-10'),
    -> ('F101','P102','Meena','12B','Chennai','Delhi','2026-08-10'),
    -> ('F102','P103','Kiran','15A','Mumbai','Bangalore','2026-08-11'),
    -> ('F103','P101','Arun','20C','Delhi','Kolkata','2026-08-12');
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM AIRLINE_BOOKING;
+----------+--------+----------+------+----------+-----------+-------------+
| FlightNo | PassID | PassName | Seat | DeptCity | ArrCity   | BookingDate |
+----------+--------+----------+------+----------+-----------+-------------+
| F101     | P101   | Arun     | 12A  | Chennai  | Delhi     | 2026-08-10  |
| F101     | P102   | Meena    | 12B  | Chennai  | Delhi     | 2026-08-10  |
| F102     | P103   | Kiran    | 15A  | Mumbai   | Bangalore | 2026-08-11  |
| F103     | P101   | Arun     | 20C  | Delhi    | Kolkata   | 2026-08-12  |
+----------+--------+----------+------+----------+-----------+-------------+
4 rows in set (0.00 sec)
```

**FUNCTIONAL DEPENDENCIES**

- PassID → PassName
- FlightNo → DeptCity, ArrCity, Date
- (FlightNo, PassID) → Seat

## CANDIDATE KEY

- (FlightNo, PassID)

Passenger information depends only on PassID, while flight information depends only on FlightNo. Hence, they are partial dependencies with respect to the composite key.

## BCNF DECOMPOSITION

MySQL 8.0 Command Line Client

```
mysql> CREATE TABLE PASSENGER (
    ->     PassID VARCHAR(10) PRIMARY KEY,
    ->     PassName VARCHAR(30)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE FLIGHT (
    ->     FlightNo VARCHAR(10) PRIMARY KEY,
    ->     DeptCity VARCHAR(30),
    ->     ArrCity VARCHAR(30),
    ->     FlightDate DATE
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE BOOKING_BCNF (
    ->     FlightNo VARCHAR(10),
    ->     PassID VARCHAR(10),
    ->     Seat VARCHAR(10),
    ->     PRIMARY KEY (FlightNo, PassID),
    ->     FOREIGN KEY (FlightNo) REFERENCES FLIGHT(FlightNo),
    ->     FOREIGN KEY (PassID) REFERENCES PASSENGER(PassID)
    -> );
Query OK, 0 rows affected (0.06 sec)

mysql> INSERT INTO PASSENGER VALUES
    -> ('P101','Arun'),
    -> ('P102','Meena'),
    -> ('P103','Kiran');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> INSERT INTO FLIGHT VALUES
    -> ('F101','Chennai','Delhi','2026-08-10'),
    -> ('F102','Mumbai','Bangalore','2026-08-11'),
    -> ('F103','Delhi','Kolkata','2026-08-12');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> INSERT INTO BOOKING_BCNF VALUES
    -> ('F101','P101','12A'),
    -> ('F101','P102','12B'),
    -> ('F102','P103','15A'),
    -> ('F103','P101','20C');
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM PASSENGER;
+--------+----------+
| PassID | PassName |
+--------+----------+
| P101   | Arun     |
| P102   | Meena    |
| P103   | Kiran    |
+--------+----------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM FLIGHT;
+----------+----------+-----------+------------+
| FlightNo | DeptCity | ArrCity   | FlightDate |
+----------+----------+-----------+------------+
| F101     | Chennai  | Delhi     | 2026-08-10 |
| F102     | Mumbai   | Bangalore | 2026-08-11 |
| F103     | Delhi    | Kolkata   | 2026-08-12 |
+----------+----------+-----------+------------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM BOOKING_BCNF;
+----------+--------+------+
| FlightNo | PassID | Seat |
+----------+--------+------+
| F101     | P101   | 12A  |
| F101     | P102   | 12B  |
| F102     | P103   | 15A  |
| F103     | P101   | 20C  |
+----------+--------+------+
4 rows in set (0.00 sec)
```

**RESULT:**

Thus, the airline booking relation was successfully normalized up to BCNF by removing the partial dependencies and separating passenger, flight, and booking information.

5) Analyze an HR system schema for a multinational company. Identify all functional and transitive dependencies and normalize to 3NF.

**AIM:**

To identify the functional and transitive dependencies in an HR system relation and normalize it up to Third Normal Form (3NF).

## GIVEN RELATION:

MySQL 8.0 Command Line Client

```
mysql> CREATE TABLE HR_EMPLOYEE (
    ->     EmpID VARCHAR(10) PRIMARY KEY,
    ->     EmpName VARCHAR(50),
    ->     DeptID VARCHAR(10),
    ->     DeptName VARCHAR(50),
    ->     ManagerID VARCHAR(10),
    ->     ManagerName VARCHAR(50),
    ->     CountryID VARCHAR(10),
    ->     CountryName VARCHAR(50),
    ->     JobID VARCHAR(10),
    ->     JobTitle VARCHAR(50),
    ->     Salary DECIMAL(10,2)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO HR_EMPLOYEE VALUES
    -> ('E101','Arun','D01','Engineering','M01','Ravi','C01','India','J01','Software Engineer',55000),
    -> ('E102','Meena','D01','Engineering','M01','Ravi','C01','India','J02','Senior Engineer',70000),
    -> ('E103','John','D02','Finance','M02','David','C02','USA','J03','Financial Analyst',65000),
    -> ('E104','Sarah','D03','Human Resources','M03','Emma','C03','UK','J04','HR Executive',60000);
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM HR_EMPLOYEE;
+-------+---------+--------+-----------------+-----------+-------------+-----------+-------------+-------+-------------------+----------+
| EmpID | EmpName | DeptID | DeptName        | ManagerID | ManagerName | CountryID | CountryName | JobID | JobTitle          | Salary   |
+-------+---------+--------+-----------------+-----------+-------------+-----------+-------------+-------+-------------------+----------+
| E101  | Arun    | D01    | Engineering     | M01       | Ravi        | C01       | India       | J01   | Software Engineer | 55000.00 |
| E102  | Meena   | D01    | Engineering     | M01       | Ravi        | C01       | India       | J02   | Senior Engineer   | 70000.00 |
| E103  | John    | D02    | Finance         | M02       | David       | C02       | USA         | J03   | Financial Analyst | 65000.00 |
| E104  | Sarah   | D03    | Human Resources | M03       | Emma        | C03       | UK          | J04   | HR Executive      | 60000.00 |
+-------+---------+--------+-----------------+-----------+-------------+-----------+-------------+-------+-------------------+----------+
4 rows in set (0.00 sec)

mysql>
```

## FUNCTIONAL DEPENDENCIES:

- EmpID → EmpName, DeptID, ManagerID, CountryID, JobID, Salary
- DeptID → DeptName, ManagerID
- ManagerID → ManagerName
- CountryID → CountryName
- JobID → JobTitle

## CANDIDATE KEY

- EmpID

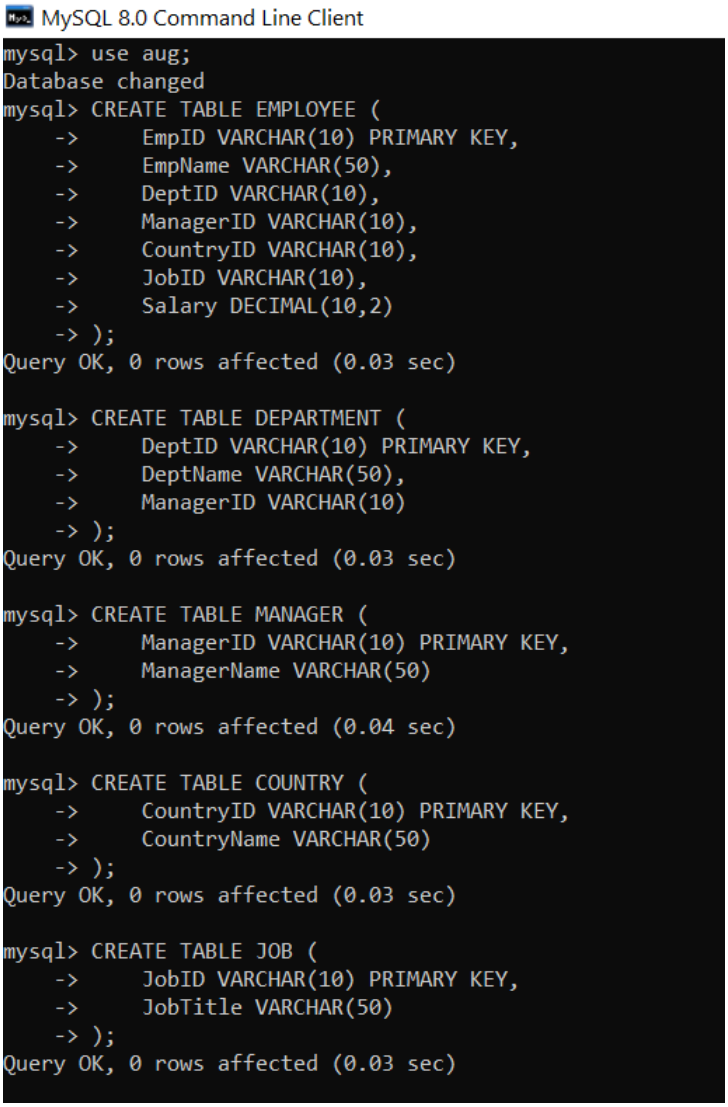
## TRANSITIVE DEPENDENCIES

The following are transitive dependencies:

- EmpID → DeptID → DeptName
- EmpID → ManagerID → ManagerName
- EmpID → CountryID → CountryName
- EmpID → JobID → JobTitle

Therefore, attributes such as DeptName, ManagerName, CountryName, and JobTitle depend indirectly on EmpID.

**3NF DECOMPOSITION**



```
mysql> INSERT INTO MANAGER VALUES
    -> ('M01','Ravi'),
    -> ('M02','David'),
    -> ('M03','Emma');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> INSERT INTO COUNTRY VALUES
    -> ('C01','India'),
    -> ('C02','USA'),
    -> ('C03','UK');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> INSERT INTO JOB VALUES
    -> ('J01','Software Engineer'),
    -> ('J02','Senior Engineer'),
    -> ('J03','Financial Analyst'),
    -> ('J04','HR Executive');
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> INSERT INTO DEPARTMENT VALUES
    -> ('D01','Engineering','M01'),
    -> ('D02','Finance','M02'),
    -> ('D03','Human Resources','M03');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> INSERT INTO EMPLOYEE VALUES
    -> ('E101','Arun','D01','M01','C01','J01',55000),
    -> ('E102','Meena','D01','M01','C01','J02',70000),
    -> ('E103','John','D02','M02','C02','J03',65000),
    -> ('E104','Sarah','D03','M03','C03','J04',60000);
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

MySQL 8.0 Command Line Client

```
mysql> SELECT * FROM EMPLOYEE;
+-------+---------+--------+-----------+-----------+-------+----------+
| EmpID | EmpName | DeptID | ManagerID | CountryID | JobID | Salary   |
+-------+---------+--------+-----------+-----------+-------+----------+
| E101  | Arun    | D01    | M01       | C01       | J01   | 55000.00 |
| E102  | Meena   | D01    | M01       | C01       | J02   | 70000.00 |
| E103  | John    | D02    | M02       | C02       | J03   | 65000.00 |
| E104  | Sarah   | D03    | M03       | C03       | J04   | 60000.00 |
+-------+---------+--------+-----------+-----------+-------+----------+
4 rows in set (0.00 sec)

mysql> SELECT * FROM DEPARTMENT;
+--------+-----------------+-----------+
| DeptID | DeptName        | ManagerID |
+--------+-----------------+-----------+
| D01    | Engineering     | M01       |
| D02    | Finance         | M02       |
| D03    | Human Resources | M03       |
+--------+-----------------+-----------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM MANAGER;
+-----------+-------------+
| ManagerID | ManagerName |
+-----------+-------------+
| M01       | Ravi        |
| M02       | David       |
| M03       | Emma        |
+-----------+-------------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM COUNTRY;
+-----------+-------------+
| CountryID | CountryName |
+-----------+-------------+
| C01       | India       |
| C02       | USA         |
| C03       | UK          |
+-----------+-------------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM JOB;
+-------+------------------+
| JobID | JobTitle         |
+-------+------------------+
| J01   | Software Engineer |
| J02   | Senior Engineer   |
| J03   | Financial Analyst |
| J04   | HR Executive      |
+-------+------------------+
```

**RESULT:**

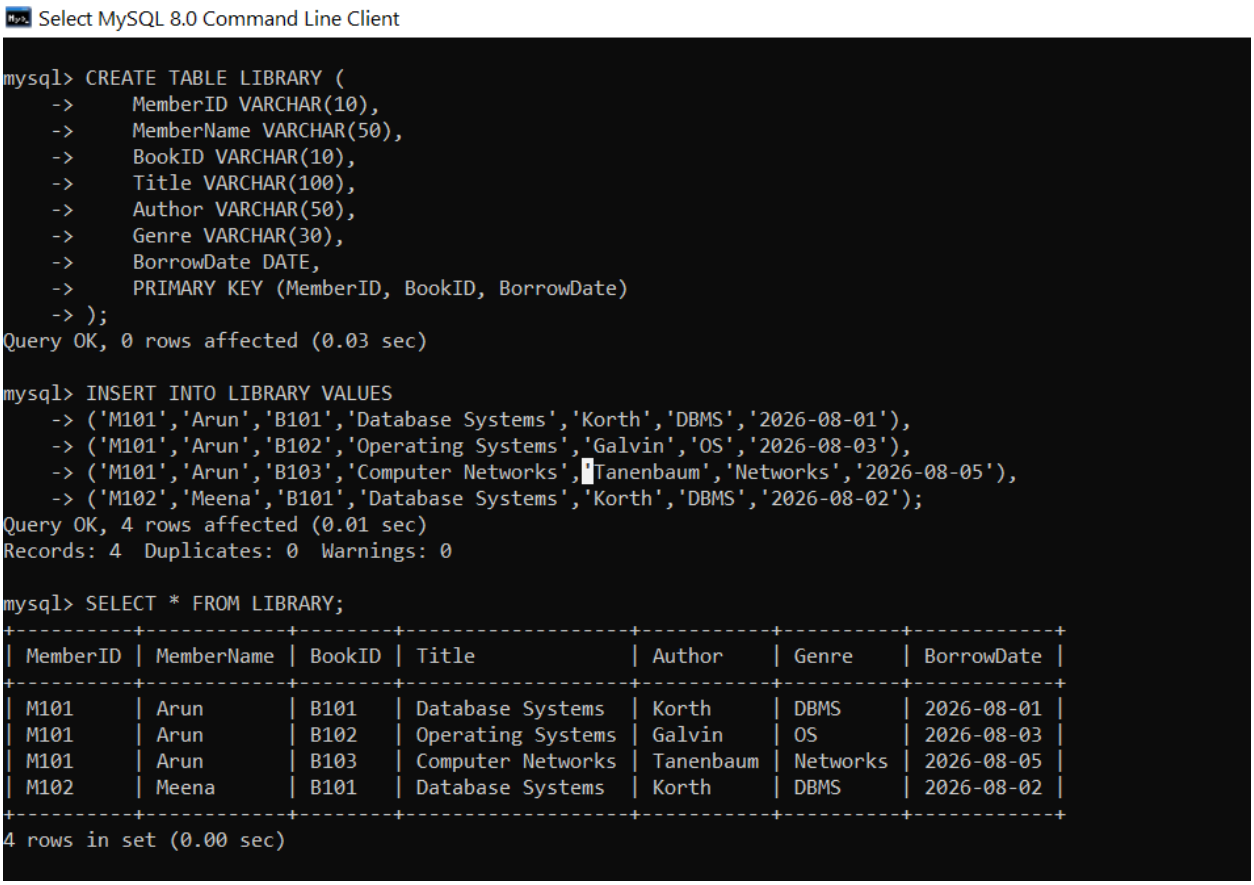
Thus, the multinational HR system relation was successfully normalized up to 3NF by identifying all functional and transitive dependencies and decomposing the relation into EMPLOYEE, DEPARTMENT, MANAGER, COUNTRY, and JOB tables. This reduces data redundancy and prevents insertion, update, and deletion anomalies.

6) Case: Library(MemberID, MemberName, BookID, Title, Author, Genre, BorrowDate). Identify MVDs and decompose to 4NF.

**AIM:**

To identify the multivalued dependencies in the Library relation and normalize it up to Fourth Normal Form (4NF).

**GIVEN RELATION:**



## MULTIVALUED DEPENDENCIES

Assume that a member can borrow **multiple books**, and a book can have multiple independent genres/authors associated with it.

The important multivalued dependencies are:

- MemberID →→ BookID
- MemberID →→ BorrowDate

This means a member may have multiple books and multiple borrowing dates independently.

## CREATE 4NF TABLES

MySQL 8.0 Command Line Client

```
mysql> CREATE TABLE MEMBER (
    ->     MemberID VARCHAR(10) PRIMARY KEY,
    ->     MemberName VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE BOOK (
    ->     BookID VARCHAR(10) PRIMARY KEY,
    ->     Title VARCHAR(100),
    ->     Author VARCHAR(50),
    ->     Genre VARCHAR(30)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE BORROW (
    ->     MemberID VARCHAR(10),
    ->     BookID VARCHAR(10),
    ->     BorrowDate DATE,
    ->     PRIMARY KEY (MemberID, BookID, BorrowDate),
    ->     FOREIGN KEY (MemberID) REFERENCES MEMBER(MemberID),
    ->     FOREIGN KEY (BookID) REFERENCES BOOK(BookID)
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql> INSERT INTO MEMBER VALUES
    -> ('M101','Arun'),
    -> ('M102','Meena');
Query OK, 2 rows affected (0.01 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> INSERT INTO BOOK VALUES
    -> ('B101','Database Systems','Korth','DBMS'),
    -> ('B102','Operating Systems','Galvin','OS'),
    -> ('B103','Computer Networks','Tanenbaum','Networks');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> INSERT INTO BORROW VALUES
    -> ('M101','B101','2026-08-01'),
    -> ('M101','B102','2026-08-03'),
    -> ('M101','B103','2026-08-05'),
    -> ('M102','B101','2026-08-02');
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM MEMBER;
+----------+------------+
| MemberID | MemberName |
+----------+------------+
| M101     | Arun       |
| M102     | Meena      |
+----------+------------+
2 rows in set (0.00 sec)

mysql> SELECT * FROM BOOK;
+--------+-------------------+-----------+----------+
| BookID | Title             | Author    | Genre    |
+--------+-------------------+-----------+----------+
| B101   | Database Systems  | Korth     | DBMS     |
| B102   | Operating Systems | Galvin    | OS       |
| B103   | Computer Networks | Tanenbaum | Networks |
+--------+-------------------+-----------+----------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM BORROW;
+----------+--------+------------+
| MemberID | BookID | BorrowDate |
+----------+--------+------------+
| M101     | B101   | 2026-08-01 |
| M102     | B101   | 2026-08-02 |
| M101     | B102   | 2026-08-03 |
| M101     | B103   | 2026-08-05 |
+----------+--------+------------+
4 rows in set (0.00 sec)
```

**RESULT:**

Thus, the Library relation was successfully decomposed into 4NF by identifying the multivalued dependencies and separating member, book, and borrowing information. This eliminates redundancy caused by independent multivalued facts and prevents modification anomalies.

7) Analyze a telecom billing schema and identify all candidate keys, primary keys, and functional dependencies before normalization.

**AIM:**

To analyze a telecom billing relation and identify all candidate keys, primary key, and functional dependencies before normalization.

**GIVEN RELATION:**

MySQL 8.0 Command Line Client

```
mysql> CREATE TABLE TELECOM_BILLING (
    ->     CustomerID VARCHAR(10),
    ->     CustomerName VARCHAR(50),
    ->     PhoneNo VARCHAR(15),
    ->     PlanID VARCHAR(10),
    ->     PlanName VARCHAR(50),
    ->     PlanType VARCHAR(30),
    ->     MonthlyCharge DECIMAL(10,2),
    ->     BillID VARCHAR(10),
    ->     BillDate DATE,
    ->     DueDate DATE,
    ->     UsageID VARCHAR(10),
    ->     CallMinutes INT,
    ->     DataUsed DECIMAL(10,2),
    ->     SMSCount INT,
    ->     BillAmount DECIMAL(10,2),
    ->
    ->     PRIMARY KEY (BillID, UsageID)
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO TELECOM_BILLING VALUES
    -> ('C101','Arun','9876543210','P101','Smart Plan','Postpaid',
    -> 499.00,'B101','2026-08-01','2026-08-15','U101',
    -> 250,12.5,100,699.00);
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> INSERT INTO TELECOM_BILLING VALUES
    -> ('C102','Meena','9876543211','P102','Premium Plan','Postpaid',
    -> 799.00,'B102','2026-08-01','2026-08-15','U102',
    -> 400,25.0,200,999.00);
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> INSERT INTO TELECOM_BILLING VALUES
    -> ('C103','Kiran','9876543212','P103','Basic Plan','Prepaid',
    -> 299.00,'B103','2026-08-01','2026-08-15','U103',
    -> 150,8.5,75,399.00);
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> INSERT INTO TELECOM_BILLING VALUES
    -> ('C101','Arun','9876543210','P101','Smart Plan','Postpaid',
    -> 499.00,'B104','2026-09-01','2026-09-15','U104',
    -> 300,15.0,120,749.00);
Query OK, 1 row affected (0.00 sec)
```

```
mysql> SELECT * FROM TELECOM_BILLING;
+------------+--------------+------------+--------+--------------+----------+--------------+--------+------------+------------+---------+------------+----------+----------+------------+
| CustomerID | CustomerName | PhoneNo    | PlanID | PlanName     | PlanType | MonthlyCharge | BillID | BillDate   | DueDate    | UsageID | CallMinutes | DataUsed | SMSCount | BillAmount |
+------------+--------------+------------+--------+--------------+----------+--------------+--------+------------+------------+---------+------------+----------+----------+------------+
| C101       | Arun         | 9876543210 | P101   | Smart Plan   | Postpaid |       499.00 | B101   | 2026-08-01 | 2026-08-15 | U101    |        250 |    12.50 |      100 |     699.00 |
| C102       | Meena        | 9876543211 | P102   | Premium Plan | Postpaid |       799.00 | B102   | 2026-08-01 | 2026-08-15 | U102    |        400 |    25.00 |      200 |     999.00 |
| C103       | Kiran        | 9876543212 | P103   | Basic Plan   | Prepaid  |       299.00 | B103   | 2026-08-01 | 2026-08-15 | U103    |        150 |     8.50 |       75 |     399.00 |
| C101       | Arun         | 9876543210 | P101   | Smart Plan   | Postpaid |       499.00 | B104   | 2026-09-01 | 2026-09-15 | U104    |        300 |    15.00 |      120 |     749.00 |
+------------+--------------+------------+--------+--------------+----------+--------------+--------+------------+------------+---------+------------+----------+----------+------------+
4 rows in set (0.00 sec)
```

**FUNCTIONAL DEPENDENCIES**

- CustomerID → CustomerName, PhoneNo, PlanID
- PhoneNo → CustomerID, CustomerName, PlanID
- PlanID → PlanName, PlanType, MonthlyCharge
- BillID → CustomerID, BillDate, DueDate, BillAmount
- UsageID → CustomerID, PhoneNo, CallMinutes, DataUsed, SMSCount
- (CustomerID, BillDate) → BillID, DueDate, BillAmount

**EXPLANATION OF FUNCTIONAL DEPENDENCIES:**

- CustomerID → CustomerName, PhoneNo, PlanID

Each customer has a unique customer ID, which determines the customer's name, phone number, and subscribed plan.

- PhoneNo → CustomerID, CustomerName, PlanID

Each registered phone number belongs to one customer.

- PlanID → PlanName, PlanType, MonthlyCharge

Each plan ID uniquely identifies the plan details and monthly charge.

- BillID → CustomerID, BillDate, DueDate, BillAmount

Each bill ID uniquely identifies a customer's bill and its billing details.

- UsageID → CustomerID, PhoneNo, CallMinutes, DataUsed, SMSCount

Each usage record contains the usage information for a particular customer.

- (CustomerID, BillDate) → BillID, DueDate, BillAmount

For a customer, a particular billing date identifies one bill.

**CANDIDATE KEYS**

The candidate keys are:

- CustomerID
- PhoneNo
- BillID
- UsageID

For the complete billing record, the key can be represented by:

- (BillID, UsageID)

depending on the assumption that a bill can contain multiple usage records.

**PRIMARY KEY**

A suitable primary key for the original relation is:

- BillID

However, if the table stores individual usage records associated with a bill, then:

- (BillID, UsageID)

can be used as the composite primary key.

**RESULT:**

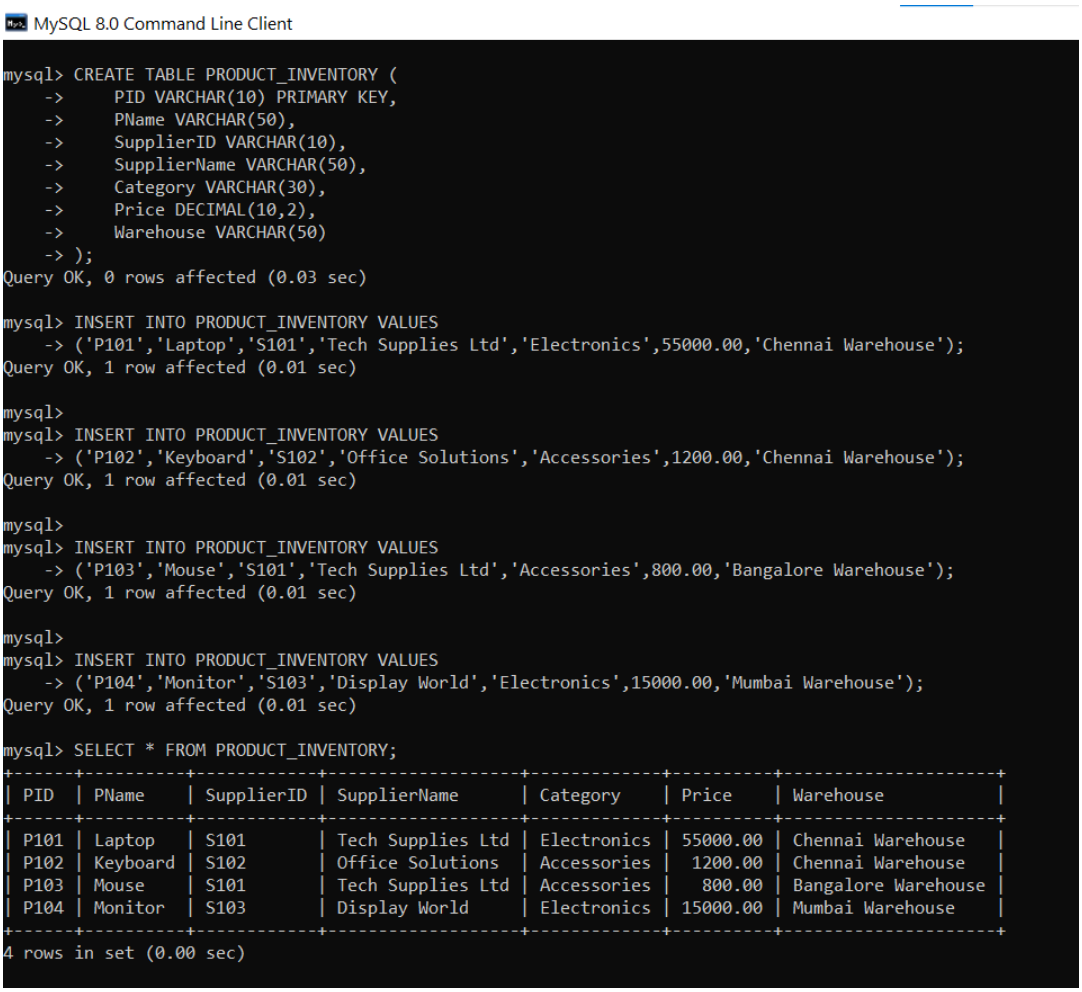
Thus, the telecom billing schema was analyzed successfully. The candidate keys, primary key, and functional dependencies were identified before normalization. These dependencies can be used in the next step to identify partial and transitive dependencies and normalize the schema up to 3NF/BCNF.

8) A retail inventory system has Product(PID, PName, SupplierID, SupplierName, Category, Price, Warehouse). Apply complete normalization.

**AIM:**

To identify the functional dependencies in the retail inventory relation and apply complete normalization up to Third Normal Form (3NF).

**GIVEN RELATION:**



## NORMALIZATION

Select MySQL 8.0 Command Line Client

```
mysql> CREATE TABLE SUPPLIER (
    ->     SupplierID VARCHAR(10) PRIMARY KEY,
    ->     SupplierName VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE PRODUCT (
    ->     PID VARCHAR(10) PRIMARY KEY,
    ->     PName VARCHAR(50),
    ->     SupplierID VARCHAR(10),
    ->     Category VARCHAR(30),
    ->     Price DECIMAL(10,2),
    ->     Warehouse VARCHAR(50),
    ->     FOREIGN KEY (SupplierID) REFERENCES SUPPLIER(SupplierID)
    -> );
Query OK, 0 rows affected (0.06 sec)

mysql> INSERT INTO SUPPLIER VALUES
    -> ('S101','Tech Supplies Ltd'),
    -> ('S102','Office Solutions'),
    -> ('S103','Display World');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> INSERT INTO PRODUCT VALUES
    -> ('P101','Laptop','S101','Electronics',55000.00,'Chennai Warehouse'),
    -> ('P102','Keyboard','S102','Accessories',1200.00,'Chennai Warehouse'),
    -> ('P103','Mouse','S101','Accessories',800.00,'Bangalore Warehouse'),
    -> ('P104','Monitor','S103','Electronics',15000.00,'Mumbai Warehouse');
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM SUPPLIER;
+------------+-------------------+
| SupplierID | SupplierName      |
+------------+-------------------+
| S101       | Tech Supplies Ltd |
| S102       | Office Solutions  |
| S103       | Display World     |
+------------+-------------------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM PRODUCT;
+------+----------+------------+-------------+----------+---------------------+
| PID  | PName    | SupplierID | Category    | Price    | Warehouse           |
+------+----------+------------+-------------+----------+---------------------+
| P101 | Laptop   | S101       | Electronics | 55000.00 | Chennai Warehouse   |
| P102 | Keyboard | S102       | Accessories |  1200.00 | Chennai Warehouse   |
| P103 | Mouse    | S101       | Accessories |   800.00 | Bangalore Warehouse |
| P104 | Monitor  | S103       | Electronics | 15000.00 | Mumbai Warehouse    |
+------+----------+------------+-------------+----------+---------------------+
4 rows in set (0.00 sec)
```

**RESULT:**

Thus, the retail inventory relation was successfully normalized up to 3NF by identifying the functional and transitive dependencies and separating supplier information into a separate SUPPLIER table. This reduces data redundancy and prevents update, insertion, and deletion anomalies.

9) Study the given poorly normalized schema for a School ERP and propose a fully normalized design up to BCNF with justification.

**AIM:**

To analyze the poorly normalized School ERP schema and decompose it into relations up to BCNF, eliminating partial and transitive dependencies and reducing redundancy.

**GIVEN RELATION:**

MySQL 8.0 Command Line Client

```
mysql> CREATE TABLE SCHOOL_ERP (
    ->     StudentID VARCHAR(10),
    ->     StudentName VARCHAR(50),
    ->     ClassID VARCHAR(10),
    ->     ClassName VARCHAR(30),
    ->     TeacherID VARCHAR(10),
    ->     TeacherName VARCHAR(50),
    ->     SubjectID VARCHAR(10),
    ->     SubjectName VARCHAR(50),
    ->     ExamID VARCHAR(10),
    ->     ExamName VARCHAR(50),
    ->     Marks INT,
    ->
    ->     PRIMARY KEY (StudentID, ExamID)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO SCHOOL_ERP VALUES
    -> ('ST101','Arun','C01','CSE-A','T01','Ravi',
    ->  'SUB01','DBMS','EX01','Internal Test 1',85);
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> INSERT INTO SCHOOL_ERP VALUES
    -> ('ST101','Arun','C01','CSE-A','T02','Meena',
    ->  'SUB02','Operating Systems','EX02','Internal Test 1',78);
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> INSERT INTO SCHOOL_ERP VALUES
    -> ('ST102','Kiran','C01','CSE-A','T01','Ravi',
    ->  'SUB01','DBMS','EX01','Internal Test 1',90);
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> INSERT INTO SCHOOL_ERP VALUES
    -> ('ST103','Meena','C02','CSE-B','T03','David',
    ->  'SUB03','Computer Networks','EX03','Internal Test 2',88);
Query OK, 1 row affected (0.01 sec)
```

```
mysql> SELECT * FROM SCHOOL_ERP;
+-----------+-------------+---------+-----------+-----------+-------------+-----------+------------------+--------+-----------------+-------+
| StudentID | StudentName | ClassID | ClassName | TeacherID | TeacherName | SubjectID | SubjectName      | ExamID | ExamName        | Marks |
+-----------+-------------+---------+-----------+-----------+-------------+-----------+------------------+--------+-----------------+-------+
| ST101     | Arun        | C01     | CSE-A     | T01       | Ravi        | SUB01     | DBMS             | EX01   | Internal Test 1 |    85 |
| ST101     | Arun        | C01     | CSE-A     | T02       | Meena       | SUB02     | Operating Systems | EX02   | Internal Test 1 |    78 |
| ST102     | Kiran       | C01     | CSE-A     | T01       | Ravi        | SUB01     | DBMS             | EX01   | Internal Test 1 |    90 |
| ST103     | Meena       | C02     | CSE-B     | T03       | David       | SUB03     | Computer Networks | EX03   | Internal Test 2 |    88 |
+-----------+-------------+---------+-----------+-----------+-------------+-----------+------------------+--------+-----------------+-------+
4 rows in set (0.00 sec)
```

**FUNCTIONAL DEPENDENCIES**

- StudentID → StudentName
- ClassID → ClassName, TeacherID
- TeacherID → TeacherName
- SubjectID → SubjectName
- ExamID → ExamName, ClassID, SubjectID
- (StudentID, ExamID) → Marks

**CANDIDATE KEY**

- (StudentID, ExamID)

A student can have multiple examinations, while a particular student and examination combination uniquely identifies the marks obtained.

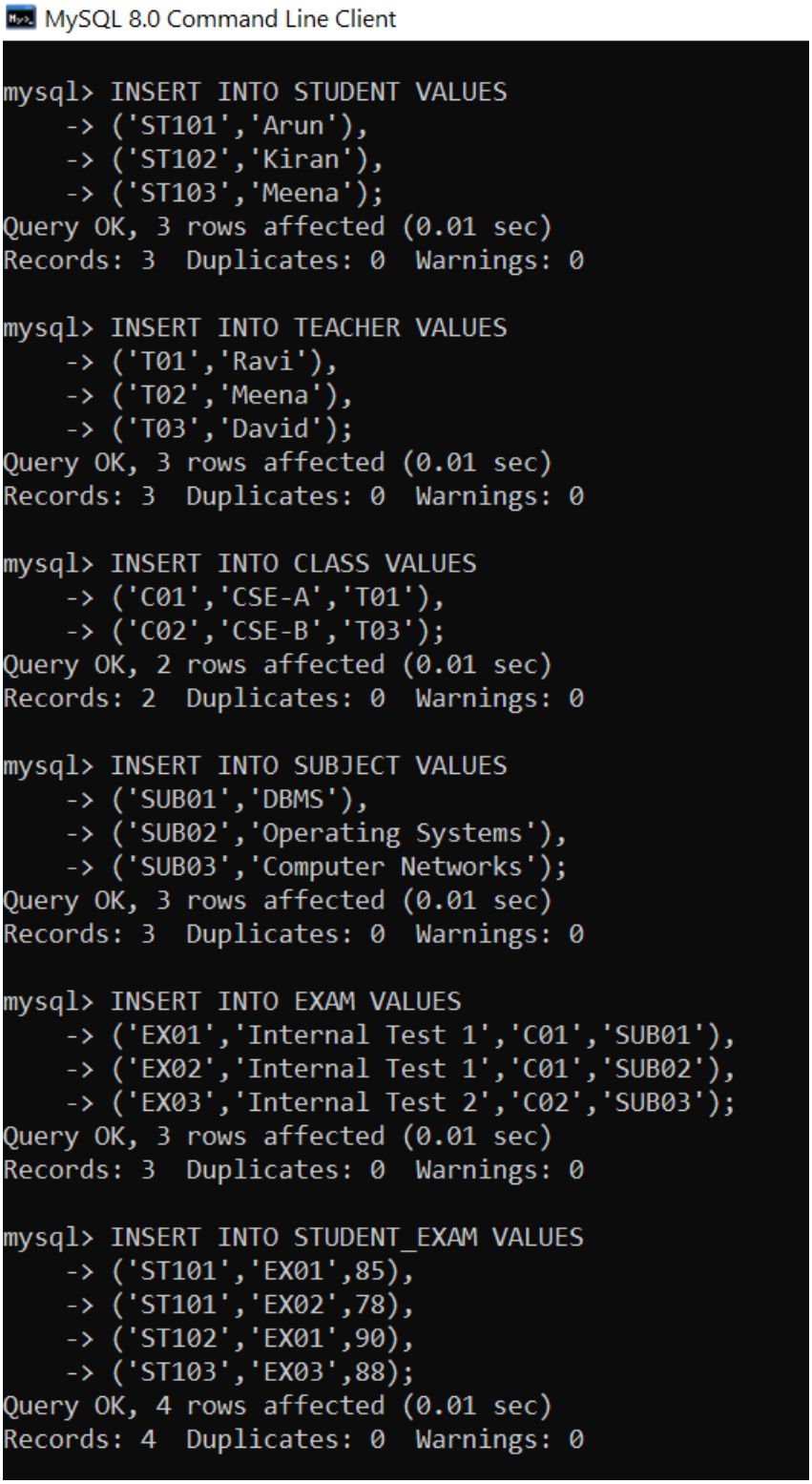
**BCNF DECOMPOSITION**

**CREATION:**

MySQL 8.0 Command Line Client

```
mysql> CREATE TABLE STUDENT (
    ->     StudentID VARCHAR(10) PRIMARY KEY,
    ->     StudentName VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE TEACHER (
    ->     TeacherID VARCHAR(10) PRIMARY KEY,
    ->     TeacherName VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql> CREATE TABLE CLASS (
    ->     ClassID VARCHAR(10) PRIMARY KEY,
    ->     ClassName VARCHAR(30),
    ->     TeacherID VARCHAR(10),
    ->     FOREIGN KEY (TeacherID) REFERENCES TEACHER(TeacherID)
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> CREATE TABLE SUBJECT (
    ->     SubjectID VARCHAR(10) PRIMARY KEY,
    ->     SubjectName VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE EXAM (
    ->     ExamID VARCHAR(10) PRIMARY KEY,
    ->     ExamName VARCHAR(50),
    ->     ClassID VARCHAR(10),
    ->     SubjectID VARCHAR(10),
    ->     FOREIGN KEY (ClassID) REFERENCES CLASS(ClassID),
    ->     FOREIGN KEY (SubjectID) REFERENCES SUBJECT(SubjectID)
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql> CREATE TABLE STUDENT_EXAM (
    ->     StudentID VARCHAR(10),
    ->     ExamID VARCHAR(10),
    ->     Marks INT,
    ->
    ->     PRIMARY KEY (StudentID, ExamID),
    ->
    ->     FOREIGN KEY (StudentID) REFERENCES STUDENT(StudentID),
    ->     FOREIGN KEY (ExamID) REFERENCES EXAM(ExamID)
    -> );
Query OK, 0 rows affected (0.04 sec)
```

**INSERTION:**


MySQL 8.0 Command Line Client
mysql> INSERT INTO STUDENT VALUES
-> ('ST101','Arun'),
-> ('ST102','Kiran'),
-> ('ST103','Meena');
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0
mysql> INSERT INTO TEACHER VALUES
-> ('T01','Ravi'),
-> ('T02','Meena'),
-> ('T03','David');
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0
mysql> INSERT INTO CLASS VALUES
-> ('C01','CSE-A','T01'),
-> ('C02','CSE-B','T03');
Query OK, 2 rows affected (0.01 sec)
Records: 2 Duplicates: 0 Warnings: 0
mysql> INSERT INTO SUBJECT VALUES
-> ('SUB01','DBMS'),
-> ('SUB02','Operating Systems'),
-> ('SUB03','Computer Networks');
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0
mysql> INSERT INTO EXAM VALUES
-> ('EX01','Internal Test 1','C01','SUB01'),
-> ('EX02','Internal Test 1','C01','SUB02'),
-> ('EX03','Internal Test 2','C02','SUB03');
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0
mysql> INSERT INTO STUDENT_EXAM VALUES
-> ('ST101','EX01',85),
-> ('ST101','EX02',78),
-> ('ST102','EX01',90),
-> ('ST103','EX03',88);
Query OK, 4 rows affected (0.01 sec)
Records: 4 Duplicates: 0 Warnings: 0

**SELECTION:**

MySQL 8.0 Command Line Client

```
mysql> SELECT * FROM STUDENT;
+-----------+-------------+
| StudentID | StudentName |
+-----------+-------------+
| ST101     | Arun        |
| ST102     | Kiran       |
| ST103     | Meena       |
+-----------+-------------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM TEACHER;
+-----------+-------------+
| TeacherID | TeacherName |
+-----------+-------------+
| T01       | Ravi        |
| T02       | Meena       |
| T03       | David       |
+-----------+-------------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM CLASS;
+---------+-----------+-----------+
| ClassID | ClassName | TeacherID |
+---------+-----------+-----------+
| C01     | CSE-A     | T01       |
| C02     | CSE-B     | T03       |
+---------+-----------+-----------+
2 rows in set (0.00 sec)

mysql> SELECT * FROM SUBJECT;
+-----------+-------------------+
| SubjectID | SubjectName       |
+-----------+-------------------+
| SUB01     | DBMS              |
| SUB02     | Operating Systems |
| SUB03     | Computer Networks |
+-----------+-------------------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM EXAM;
+--------+-----------------+---------+-----------+
| ExamID | ExamName        | ClassID | SubjectID |
+--------+-----------------+---------+-----------+
| EX01   | Internal Test 1 | C01     | SUB01     |
| EX02   | Internal Test 1 | C01     | SUB02     |
| EX03   | Internal Test 2 | C02     | SUB03     |
+--------+-----------------+---------+-----------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM STUDENT_EXAM;
+-----------+--------+-------+
| StudentID | ExamID | Marks |
+-----------+--------+-------+
| ST101     | EX01   |    85 |
| ST101     | EX02   |    78 |
| ST102     | EX01   |    90 |
| ST103     | EX03   |    88 |
+-----------+--------+-------+
4 rows in set (0.00 sec)
```

**RESULT:**

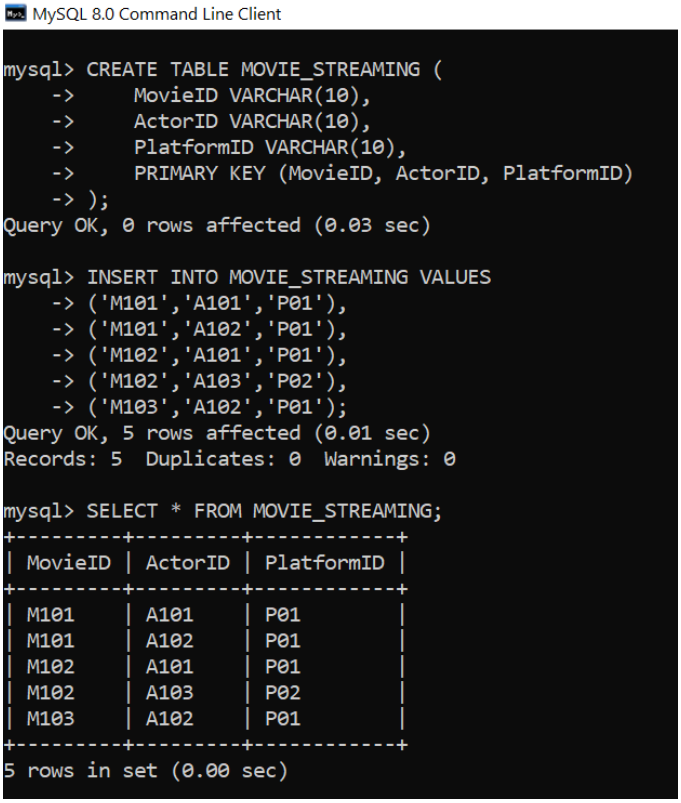
Thus, the poorly normalized School ERP schema was successfully analyzed and decomposed into BCNF relations. The decomposition removes partial and transitive dependencies, ensures that every determinant is a candidate key, and reduces redundancy while maintaining the relationships between students, classes, teachers, subjects, examinations, and marks.

10) Given a movie streaming platform schema, identify join dependencies and decompose to 5NF with lossless-join verification.

**AIM:**

To identify the join dependencies (JDs) in a movie streaming platform relation and decompose it into 5NF while verifying that the decomposition is lossless.

**GIVEN RELATION:**



## JOIN DEPENDENCIES

- *(MovieID, ActorID),
- (MovieID, PlatformID),
- (ActorID, PlatformID)

## 5NF DECOMPOSITION

MySQL 8.0 Command Line Client

```
mysql> CREATE TABLE MOVIE_ACTOR (
    ->     MovieID VARCHAR(10),
    ->     ActorID VARCHAR(10),
    ->     PRIMARY KEY (MovieID, ActorID)
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql> CREATE TABLE MOVIE_PLATFORM (
    ->     MovieID VARCHAR(10),
    ->     PlatformID VARCHAR(10),
    ->     PRIMARY KEY (MovieID, PlatformID)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE ACTOR_PLATFORM (
    ->     ActorID VARCHAR(10),
    ->     PlatformID VARCHAR(10),
    ->     PRIMARY KEY (ActorID, PlatformID)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO MOVIE_ACTOR VALUES
    -> ('M101','A101'),
    -> ('M101','A102'),
    -> ('M102','A101'),
    -> ('M102','A103'),
    -> ('M103','A102');
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> INSERT INTO MOVIE_PLATFORM VALUES
    -> ('M101','P01'),
    -> ('M102','P01'),
    -> ('M102','P02'),
    -> ('M103','P01');
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> INSERT INTO ACTOR_PLATFORM VALUES
    -> ('A101','P01'),
    -> ('A102','P01'),
    -> ('A103','P02');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM MOVIE_ACTOR;
+---------+---------+
| MovieID | ActorID |
+---------+---------+
| M101    | A101    |
| M101    | A102    |
| M102    | A101    |
| M102    | A103    |
| M103    | A102    |
+---------+---------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM MOVIE_PLATFORM;
+---------+------------+
| MovieID | PlatformID |
+---------+------------+
| M101    | P01        |
| M102    | P01        |
| M102    | P02        |
| M103    | P01        |
+---------+------------+
4 rows in set (0.00 sec)

mysql> SELECT * FROM ACTOR_PLATFORM;
+---------+------------+
| ActorID | PlatformID |
+---------+------------+
| A101    | P01        |
| A102    | P01        |
| A103    | P02        |
+---------+------------+
3 rows in set (0.00 sec)
```

## LOSSLESS-JOIN VERIFICATION

```
mysql> SELECT MA.MovieID,
    ->        MA.ActorID,
    ->        MP.PlatformID
    -> FROM MOVIE_ACTOR MA
    -> JOIN MOVIE_PLATFORM MP
    ->     ON MA.MovieID = MP.MovieID
    -> JOIN ACTOR_PLATFORM AP
    ->     ON MA.ActorID = AP.ActorID
    ->    AND MP.PlatformID = AP.PlatformID;
+---------+---------+------------+
| MovieID | ActorID | PlatformID |
+---------+---------+------------+
| M101    | A102    | P01        |
| M101    | A101    | P01        |
| M102    | A101    | P01        |
| M102    | A103    | P02        |
| M103    | A102    | P01        |
+---------+---------+------------+
5 rows in set (0.01 sec)
```

Thus,

MOVIE_STREAMING =

MOVIE_ACTOR

⋈ MOVIE_PLATFORM

⋈ ACTOR_PLATFORM

**RESULT:**

Thus, the Movie Streaming relation was successfully decomposed into 5NF by identifying the non-trivial join dependency and separating the movie–actor, movie–platform, and actor–platform relationships. The original relation can be reconstructed through natural joins, confirming that the decomposition is lossless.